Engineering Operations Policy

Document Intelligence POC — Sample Document

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1. Purpose and Scope

This policy defines the minimum controls for deploying software services into the production environment. It applies to application developers, release managers, platform engineers, and incident commanders who participate in production changes.

The policy covers standard deployments, emergency deployments, rollback decisions, monitoring requirements, and post-deployment review. It does not replace security standards, data-handling requirements, or service-specific runbooks.

2. Core Principles

Production changes must be traceable to an approved change record. A deployment must have a named owner, a documented validation plan, and a rollback procedure appropriate to the risk of the change.

Teams should prefer small, reversible changes. A successful build or passing unit tests alone does not establish that a service is safe to deploy. The release owner must consider integration tests, configuration changes, database compatibility, and operational readiness.

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3. Standard Production Deployment

3.1 Required approvals

The release manager must approve every standard production deployment before execution. The service owner must confirm that the change is ready and that the validation and rollback plans are documented. For changes that modify authentication, authorization, encryption, or sensitive-data handling, a security reviewer must also approve the change.

Approval must be recorded in the change-management system. A message in a chat channel can support coordination, but it does not replace the required approval record.

3.2 Pre-deployment checklist

Before deployment, the release owner must verify that automated tests have passed, the artifact corresponds to the reviewed source revision, required configuration is available, and dashboards and alerts are operational. Database migrations must be backward-compatible with the currently running application unless a separately approved migration plan states otherwise.

The team must identify the expected impact, the monitoring signals that indicate success, and the conditions that trigger rollback. If any mandatory check fails, the deployment must be paused until the failure is resolved or an authorized exception is recorded.

3.3 Deployment execution

Deployments should proceed in stages when the platform supports staged rollout. After each stage, the release owner should inspect error rate, latency, saturation, and key business metrics before increasing traffic. The release record must include the start time, deployed version, responsible operator, and final outcome.

4. Emergency Deployment

An emergency deployment is permitted only when delaying a change creates a material risk to service availability, data integrity, or security. The incident commander must authorize an emergency deployment before execution whenever circumstances permit. If immediate action is required to prevent active harm, the on-call engineer may act first and must notify the incident commander as soon as practical.

Emergency changes must still have a named operator, a concise risk assessment, a validation step, and a rollback or containment plan. The team must record why the standard approval path was not followed and link the change to the incident record.

4.1 Post-emergency review

The service owner and incident commander must review the emergency change within two business days. The review should confirm the actual impact, identify follow-up work, and document any temporary controls that remain in place. The team must convert unresolved risks into tracked actions with owners and due dates.

5. Rollback and Recovery

The release owner must initiate rollback when agreed success criteria fail, when customer impact exceeds the approved threshold, or when a critical monitoring signal indicates that the service is unstable. The incident commander may direct rollback during an active incident.

A rollback must not be treated as complete until service health is verified. If reverting the application would make the data store inconsistent or incompatible, the team must use the documented recovery procedure instead of blindly redeploying the previous version.

After rollback, the operator must record the trigger, timeline, observed impact, recovery action, and verification results in the change record. A follow-up review is required for any rollback that affects customers or causes a material service interruption.

6. Monitoring and Operational Readiness

Before a production deployment begins, the service must have usable health checks and monitoring for request errors, latency, resource saturation, and any critical business operation. Alerts must route to a team that has an active on-call process. A dashboard without an owner or response procedure is not sufficient operational readiness.

During rollout, the release owner must compare observed behavior with the pre-deployment baseline. A brief increase in latency may be acceptable if it remains within the approved threshold and returns to baseline. Sustained error growth, failed health checks, or unexpected data changes require the rollout to stop while the team investigates.

6.1 Incident communication

When a deployment causes or may be contributing to an incident, the operator must notify the incident channel and provide the change record, affected service, current impact, and immediate mitigation. Updates should distinguish confirmed facts from hypotheses. Do not claim recovery until monitoring and a functional check support that conclusion.

7. Security and Access Controls

Production access must use approved individual identities and least-privilege permissions. Shared credentials must not be used for routine deployment. Secrets must be obtained from the approved secret-management system and must not be committed to source control or copied into deployment logs.

Changes to access control, authentication, authorization, encryption, or sensitive-data handling require security review under the standard deployment process. Emergency changes involving these areas must be reviewed retrospectively as part of the post-emergency review.

8. Records, Exceptions, and Ownership

The change record is the authoritative record for standard deployment approvals, validation evidence, deployment timing, operator identity, outcome, and rollback details. Links to build logs, test reports, dashboards, and incident records should be included when they are relevant and accessible to the reviewing team.

Exceptions must state the requirement being waived, the reason, the risk, the compensating controls, the approving person, and the expiration date. An exception does not permanently amend this policy. Expired exceptions must not be treated as authorization for future deployments.

9. Responsibilities

Release manager: Approves standard production deployments and verifies that mandatory checks and approvals are recorded.

Service owner: Confirms service readiness, owns validation and rollback plans, and participates in post-deployment or post-emergency reviews.

Security reviewer: Reviews standard changes affecting authentication, authorization, encryption, or sensitive-data handling.

Incident commander: Authorizes emergency deployments when feasible, coordinates response during active incidents, and may direct rollback.

On-call engineer: Executes authorized changes, monitors service health, records evidence, and escalates unexpected impact.

10. Quick Reference

Standard deployment: release manager approval is mandatory before execution. Emergency deployment: incident commander authorization is required before execution when circumstances permit; urgent action to prevent active harm may precede notification. Rollback: trigger when agreed success criteria fail or critical instability is observed, then verify service health and record the outcome.

End of fictional sample policy.